

Sri Lanka Institute of Information Technology



Chronic Kidney Disease Prediction

Final Group Report

Problem Domain: Healthcare

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1. Introduction and Problem Statement

Chronic Kidney Disease (CKD) is a major global health challenge, affecting approximately 10% of the world's population according to the World Health Organization (WHO). A significant issue with this disease is that it can develop for years with only mild or no symptoms, often going undetected until serious complications—such as kidney failure requiring dialysis—have already occurred. This delay in diagnosis severely impacts patients' quality of life and places a heavy burden on healthcare systems, with high costs associated with late-stage treatments like dialysis.

The core problem we address is the need for early and accessible identification of individuals at high risk of developing CKD. Traditional diagnostic methods often rely on invasive or costly tests, such as glomerular filtration rate (GFR) measurements or creatinine levels, which may not be accessible, especially in rural areas. This is where Artificial Intelligence (AI) and Machine Learning (ML) can play a transformative role.

Therefore, this project aims to develop a machine learning model for the early prediction of CKD and the need for dialysis. Our solution, named RiskGuard AI, is designed not as a standalone diagnostic tool, but as an intelligent assistant for healthcare professionals. By analyzing key health indicators, the model will calculate a personalized risk score to help flag at-risk individuals for further medical evaluation. The ultimate goal is to enable earlier intervention, prevent kidney failure, reduce dialysis costs, improve patient outcomes, and contribute to more efficient healthcare delivery.

2. Dataset Description

2.1 Dataset Source & Overview: The primary dataset for this project is the "Kidney Disease Risk Dataset," sourced from Kaggle, accessible at <https://www.kaggle.com/datasets/miadul/kidney-disease-risk-dataset>. This dataset comprises synthetic yet medically realistic patient profiles, designed for binary classification tasks. It aims to predict the presence (1) or absence (0) of Chronic Kidney Disease (CKD) and the need for dialysis, aligning perfectly with the objectives of RiskGuard AI.

2.2 Key Characteristics:

- **Size:** The dataset contains 1,985 records, exceeding the project's minimum requirement of 1,000 records, providing a robust foundation for model training and validation.
- **Features:** It includes 9 features (surpassing the 6-feature minimum), encompassing well-established CKD risk factors:
 - Age (numerical: patient age in years),
 - Creatinine_Level (numerical: serum creatinine in mg/dL),
 - BUN (numerical: blood urea nitrogen in mg/dL),
 - Diabetes (binary: 0 = No, 1 = Yes),
 - Hypertension (binary: 0 = No, 1 = Yes),
 - GFR (numerical: glomerular filtration rate in mL/min/1.73m²),
 - Urine_Output (numerical: daily urine volume in mL).
- **Target Variables:** CKD_Status (0 = No CKD, 1 = CKD) and Dialysis_Needed (0 = No, 1 = Yes).
- The original dataset contained 9 medically relevant features (surpassing the 6-feature minimum required for the project).
- However, after preprocessing and applying PCA for dimensionality reduction, the final dataset used for model training included 3 principal components (PC1, PC2, PC3), which retained approximately 85% of the original variance while improving model efficiency.

2.3 Relevance and Justification: This dataset is highly relevant as it incorporates critical clinical markers for CKD prediction, such as GFR, creatinine levels, and BUN, which are standard in nephrology for assessing kidney function. The dataset's size (1,985 records) supports robust model development and validation. Although synthetic and based on medical guidelines, it enables the exploration of fairness across subgroups (e.g., age, diabetes status), making it suitable for this study's ethical and bias-focused objectives.

2.4 Potential Limitations and Biases:

- **Class Imbalance:** A moderate imbalance exists in CKD_Status (approximately 55% non-CKD vs. 45% CKD), which was addressed during preprocessing to mitigate bias toward the majority class.
- **Inherent Biases:** As a synthetic dataset, it may not fully reflect real-world variabilities, such as socioeconomic factors or ethnic-specific GFR norms. However, it allows for controlled analysis of biases in features like age or hypertension.

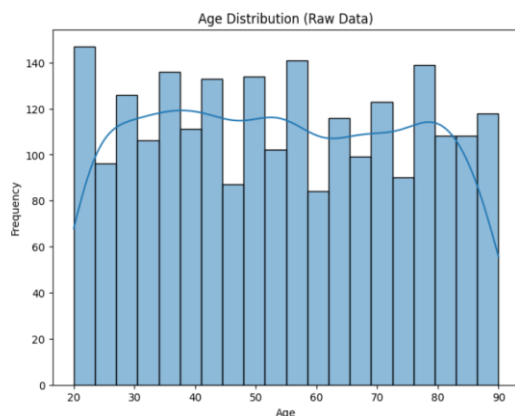
2.5 Ethical Considerations: The use of anonymized synthetic data ensures confidentiality and eliminates personally identifiable information, adhering to ethical standards for health-related research. The dataset's artificial nature also avoids privacy concerns associated with real patient data.

3. Preprocessing & Exploratory Data Analysis (EDA)

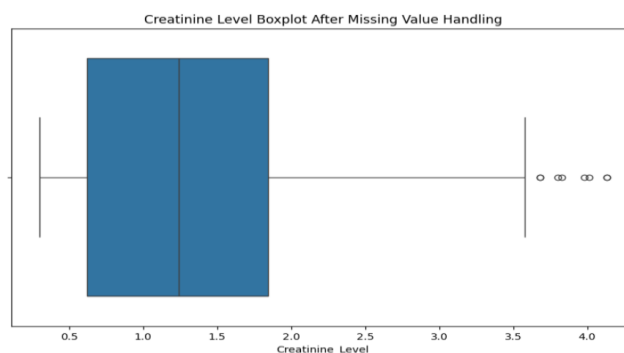
This section details the steps taken to clean the kidney disease dataset, handle missing values, normalize features, reduce dimensionality, and explore patterns. The workflow was implemented using a collaborative pipeline in Google Colab notebooks, with each member focusing on a specific preprocessing task.

3.1 Data Cleaning (Missing & Duplicate Handling)

- **Goal:** Ensure dataset has no missing or duplicate records.
- **Key Actions:**
 - Checked for missing values across all columns.
 - Filled numeric missing values with **mean** (e.g., Urine_Output, GFR) and categorical missing values with **mode** (Diabetes, Hypertension, CKD_Status, Dialysis_Needed).
 - Checked for duplicates using `drop_duplicates()` (none found).
 - Saved cleaned dataset: `dataset_after_missing_values.csv`.
- **Visualization:**
 - Histograms for numerical features, e.g., Age distribution.
 - Boxplots for features like `Creatinine_Level` to observe effects of imputation.



```
Missing values per column:
Age          0
Creatinine_Level  0
BUN          0
Diabetes      0
Hypertension  0
GFR          0
Urine_Output  0
CKD_Status   0
Dialysis_Needed  0
dtype: int64
Missing values after handling:
Age          0
Creatinine_Level  0
BUN          0
Diabetes      0
Hypertension  0
GFR          0
Urine_Output  0
CKD_Status   0
Dialysis_Needed  0
dtype: int64
```



3.2 Encoding Categorical Variables

- **Goal:** Convert categorical features into numeric format suitable for ML models.

- **Techniques Used:**

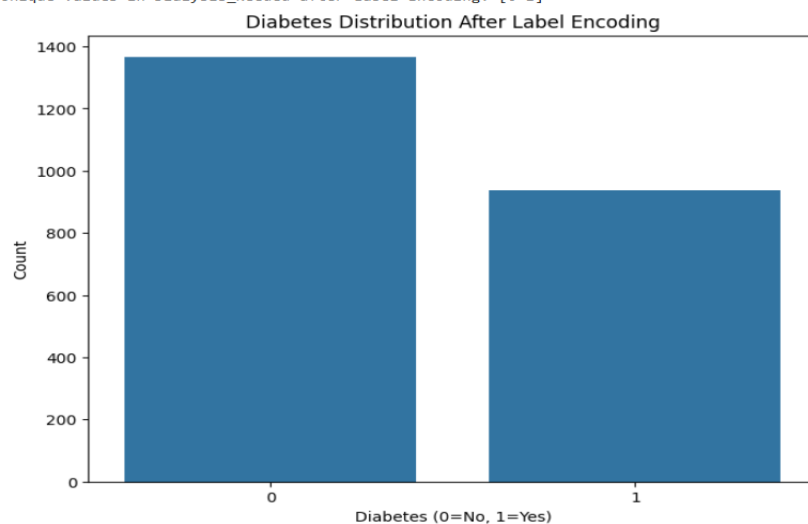
- **Label Encoding:** Applied to binary features (Diabetes, Hypertension, CKD_Status, Dialysis_Needed).
- **One-Hot Encoding:** Applied for consistency, creating dummy variables for categorical columns while dropping the first level to avoid multicollinearity.

- **Output:** dataset_after_onehot_encoding.csv.

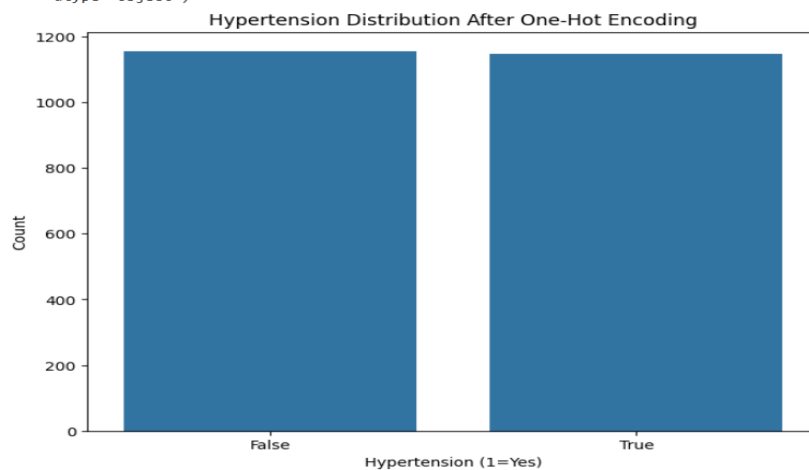
- **Visualizations:**

- Countplots for encoded features (e.g., Diabetes, Hypertension) to observe distribution changes.

```
Unique values in Diabetes after Label Encoding: [0 1]
Unique values in Hypertension after Label Encoding: [1 0]
Unique values in CKD_Status after Label Encoding: [1 0]
Unique values in Dialysis_Needed after Label Encoding: [0 1]
```



```
Columns after One-Hot Encoding: Index(['Age', 'Creatinine_Level', 'BUN', 'GFR', 'Urine_Output', 'Diabetes_1',
    'Hypertension_1', 'CKD_Status_1', 'Dialysis_Needed_1'],
    dtype='object')
```

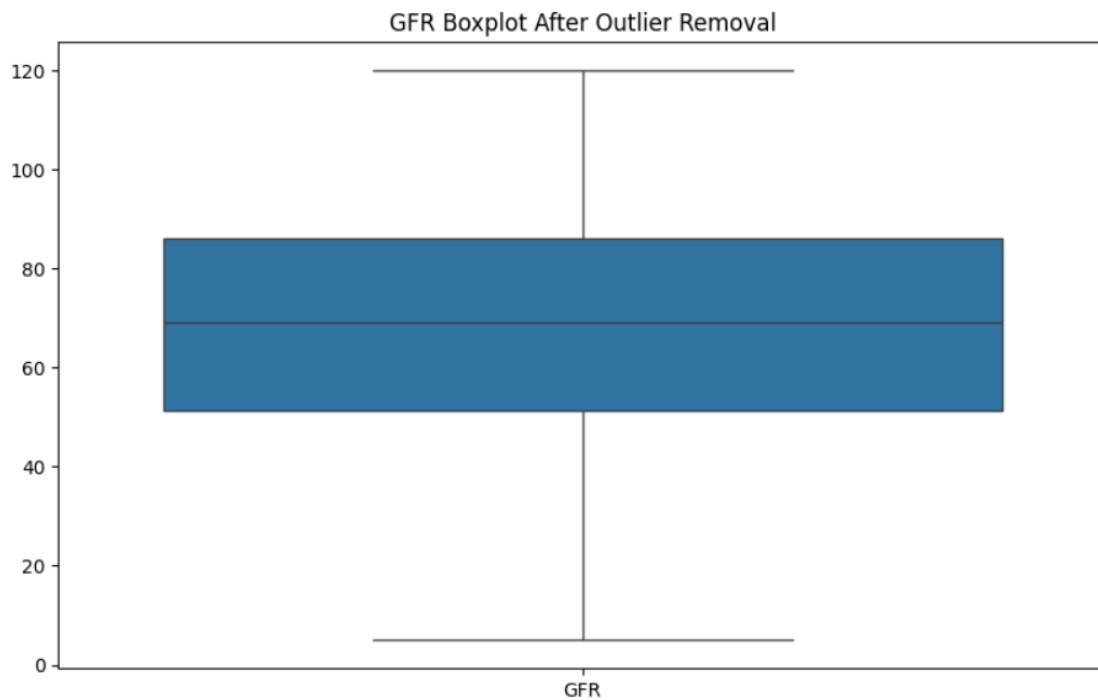


3.3 Outlier Handling

- **Goal:** Remove extreme values that could bias model training.
- **Techniques:**
 - **Z-Score Method:** Removed rows with Z-scores > 3 for numerical columns (Age, Creatinine_Level, BUN, GFR, Urine_Output).
- **Output:** dataset_after_outliers.csv.
- **Visualizations:**
 - Boxplots before and after outlier removal (e.g., GFR) to verify reduction in extreme values.

Original shape: (2304, 9)

Shape after outlier removal: (2285, 9)



3.4 Feature Scaling / Normalization

- **Goal:** Scale numeric features for fair comparison across models.

- **Technique:** MinMaxScaler (normalized to [0,1]):

$$x' = \frac{x - \min(x)}{\max(x) - \min(x)}$$

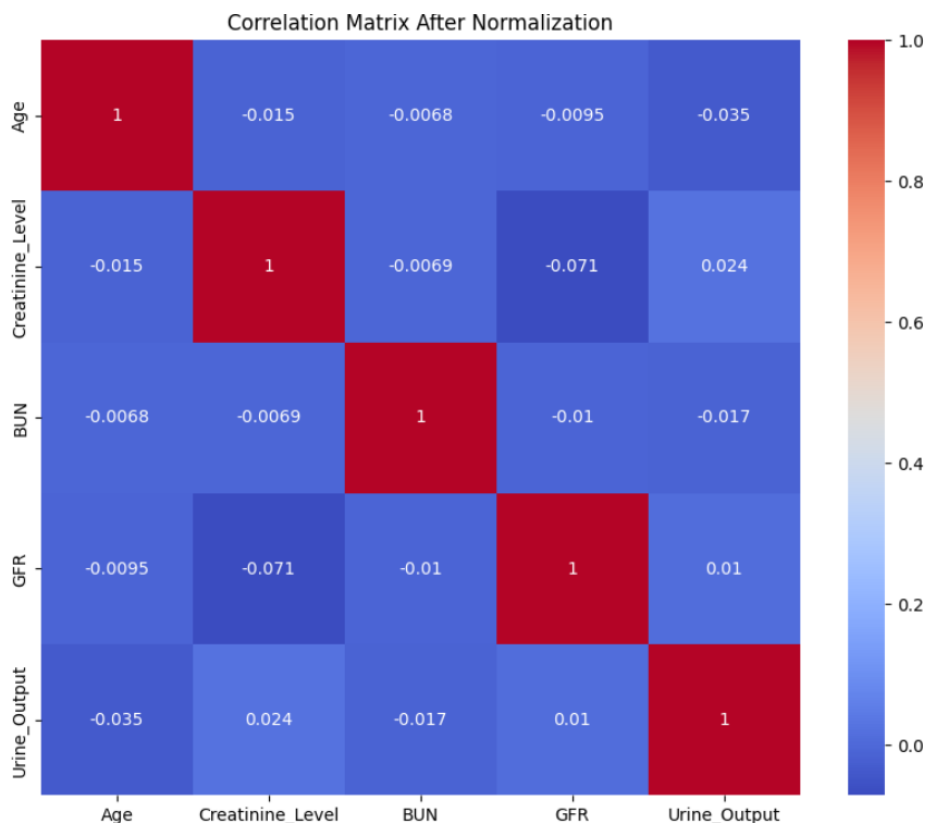
- **Columns Scaled:** Age, Creatinine_Level, BUN, GFR, Urine_Output.

- **Output:** dataset_after_normalization.csv.

- **Visualizations:**

- Correlation matrix heatmaps for normalized features.
- Histograms and boxplots showing distributions post-normalization.

	Age	Creatinine_Level	BUN	GFR	Urine_Output
count	2285.000000	2285.000000	2285.000000	2285.000000	2285.000000
mean	0.488471	0.304338	0.302614	0.555576	0.450455
std	0.292922	0.236663	0.227859	0.214530	0.182227
min	0.000000	0.000000	0.000000	0.000000	0.000000
25%	0.228571	0.097561	0.108168	0.402609	0.323891
50%	0.485714	0.283537	0.291391	0.557391	0.445024
75%	0.742857	0.469512	0.461369	0.706087	0.572121
max	1.000000	1.000000	1.000000	1.000000	1.000000



3.5. Feature Selection

- **Goal:** Identify the most relevant features for CKD prediction while reducing dimensionality.

- **Techniques:**

1. **VarianceThreshold:** Removed low-variance features (`threshold=0.1`).
 - Selected features saved in `dataset_after_variance_threshold.csv`.
2. **SelectKBest (ANOVA F-score):** Chose top 5 features most predictive of `CKD_Status_1`.
 - Key selected features included `Creatinine_Level`, `GFR`, etc.
 - Saved dataset: `dataset_after_feature_selection.csv`.

- **Visualizations:**

- Correlation heatmap of top features.
- Barplot showing top 10 ANOVA F-score features.
- Histograms and boxplots for selected features versus `CKD_Status_1`.

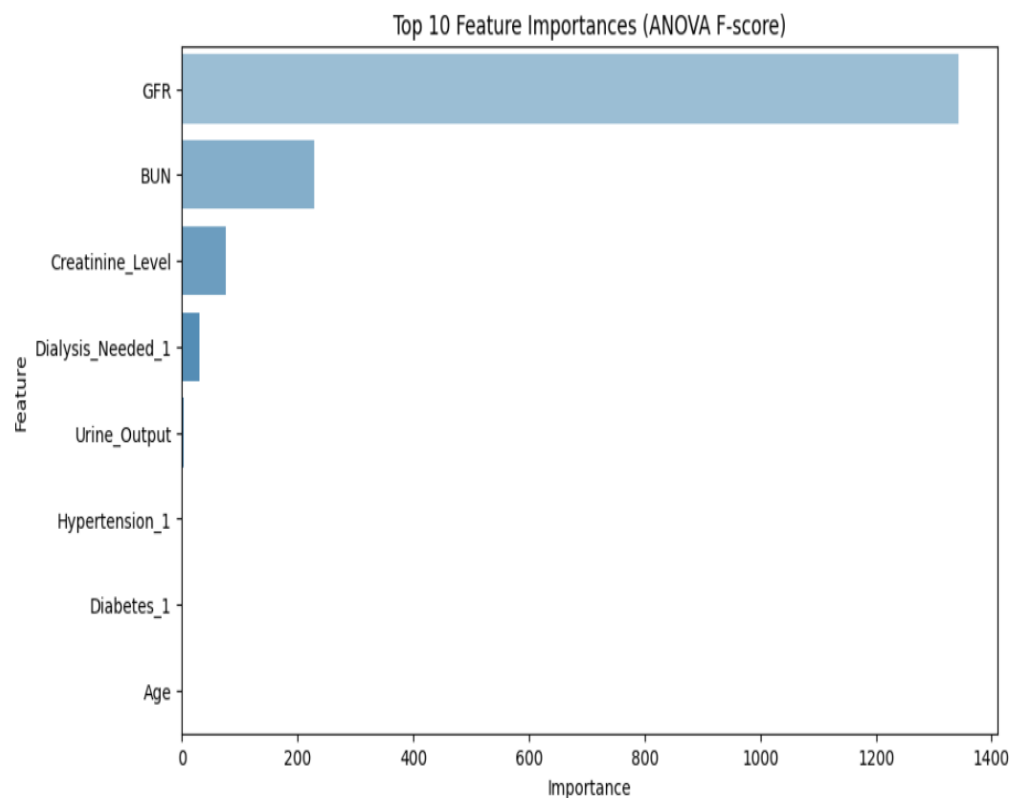
```
Selected features after VarianceThreshold: ['Diabetes_1', 'Hypertension_1']
```

```
Selected features after SelectKBest: ['Creatinine_Level', 'BUN', 'GFR', 'Urine_Output', 'Dialysis_Needed_1']
```

```
/tmp/ipython-input-62986919.py:59: FutureWarning:
```

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `y` variable to `hue` and set `legend=False` for the same effect

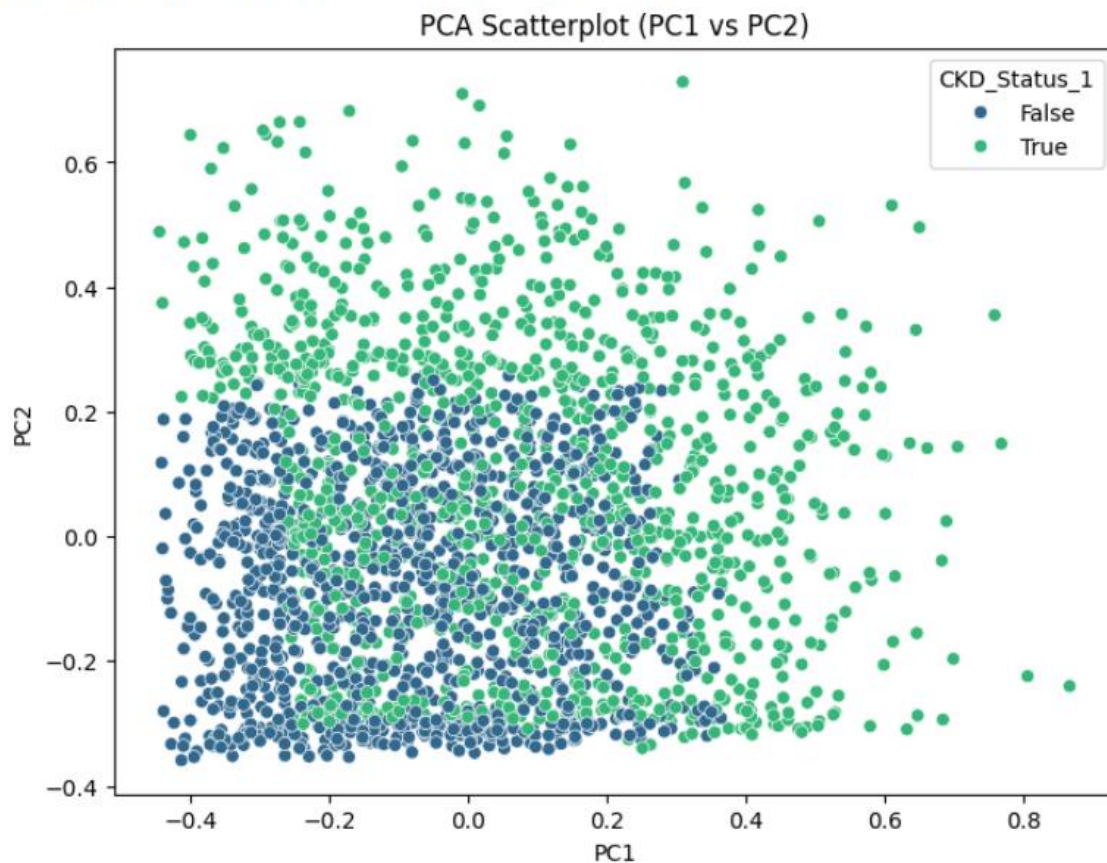
```
sns.barplot(x='Importance', y='Feature', data=importance_df, palette='Blues_d')
```



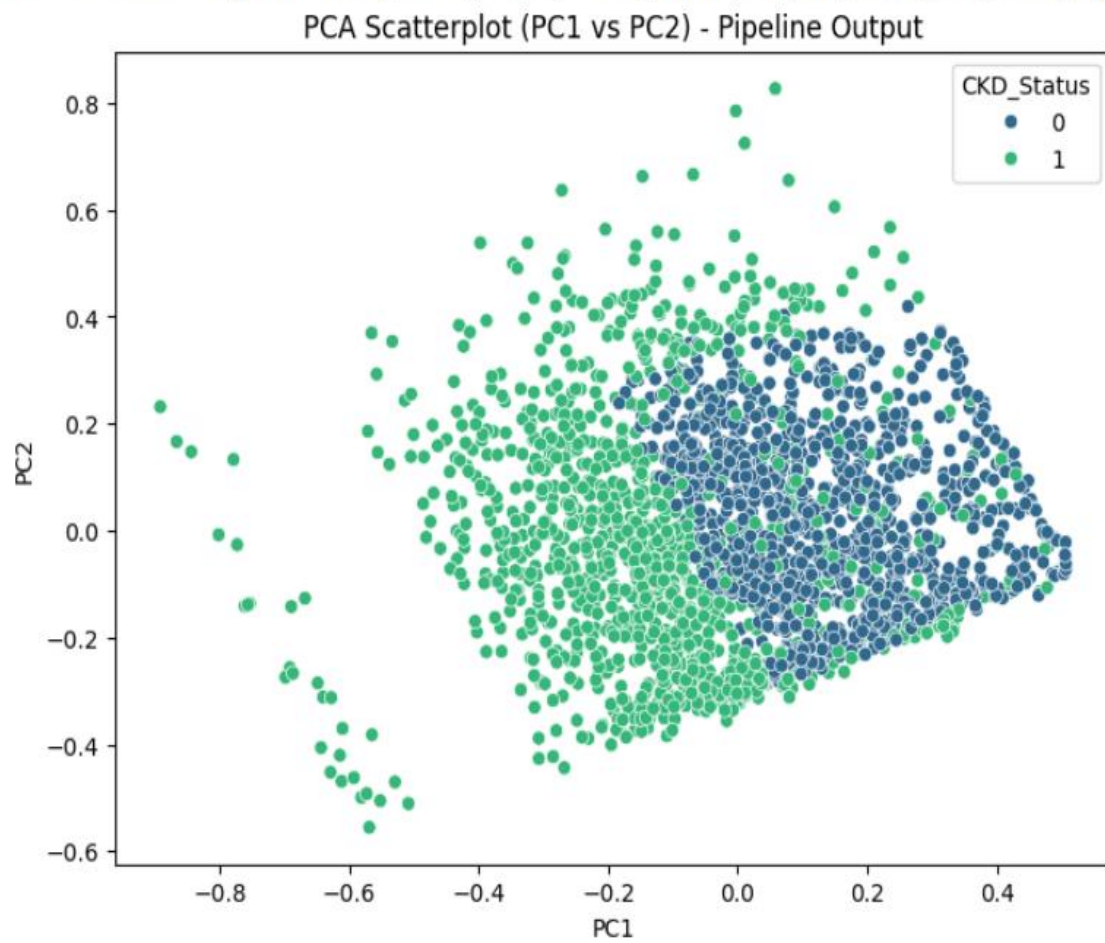
3.6 Dimensionality Reduction (PCA)

- **Goal:** Reduce feature space for visualization and efficiency while retaining maximum variance.
- **Technique:** PCA applied with 2 components.
- **Output:** `dataset_after_pca.csv`.
- **Visualizations:**
 - Scatterplot of first two principal components (PC1 vs PC2) colored by `CKD_Status_1`.
 - Explained variance ratio of PCA components to quantify information retention.

Explained variance ratio (2 PCs): [0.28616168 0.25923482]



Selected features: ['Creatinine_Level', 'BUN', 'GFR', 'Urine_Output', 'Dialysis_Needed_1']



4. Model Design and Implementation

We selected a diverse set of models to compare performance, focusing on interpretability and accuracy for healthcare applications.

4.1 Model Selection and Justification

We chose six models as follows:

- **Logistic Regression:** A baseline, interpretable model that provides coefficients for feature importance.
- **Decision Tree:** Handles non-linear relationships and provides rule-based decisions for interpretability.
- **SVM (RBF):** Effective for binary classification and margin maximization, particularly suitable for CKD detection.
- **Random Forest:** An ensemble method that improves accuracy by aggregating multiple decision trees.
- **XGBoost:** A gradient boosting model that captures complex feature interactions and enhances predictive performance.
- **MLP Neural Network:** Captures complex non-linear relationships, suitable for modeling intricate CKD patterns.
- **PCA (Variance):** Applied as a preprocessing step to reduce dimensionality while retaining ~85% of the variance, improving model efficiency.

This combination allows a comparison of models balancing **accuracy, interpretability, and computational efficiency** for CKD prediction.

4.2 Individual Model Implementation

- **Logistic Regression:** Tuned with `C=0.1` and `solver='liblinear'` to achieve optimal accuracy while maintaining interpretability.
- **Decision Tree:** Tuned with `max_depth=10`, `min_samples_leaf=5`, and `criterion='gini'` to prevent overfitting and ensure stable performance.
- **SVM (RBF):** Tuned with `kernel='rbf'`, `C=10`, and `gamma='auto'` for optimal margin separation.
- **Random Forest:** Tuned with `n_estimators=50`, `max_depth=20`, and `min_samples_split=2` to balance accuracy and overfitting control.
- **XGBoost:** Tuned with `n_estimators=300`, `max_depth=3`, `learning_rate=0.1`, `colsample_bytree=1.0`, and `subsample=1.0` for high predictive performance.
- **MLP Neural Network:** Tuned with `hidden_layer_sizes=(50,50)`, `activation='relu'`, and `alpha=0.01` to capture complex non-linear patterns.
- **PCA (Variance):** Applied to reduce dimensionality and retain ~85% of variance, enhancing computational efficiency for all models.

5. Evaluation and Comparison

We evaluated models on the test set using appropriate metrics for imbalanced healthcare data.

5.1 Evaluation Framework We used stratified 5-fold cross-validation to ensure balanced class representation; metrics included Accuracy, Precision, Recall, F1-Score, and AUC-ROC, with a priority on Recall to minimize false negatives in CKD detection.

5.2 Individual Model Performance

Model	Accuracy	Precision	Recall	F1-Score	AUC
Logistic Regression	0.8330	0.83	0.83	0.83	0.9120
SVM (RBF)	0.9761	0.98	0.98	0.98	0.9981
Decision Tree	0.9631	0.96	0.96	0.96	0.9809
Random Forest	0.9718	0.97	0.97	0.97	0.9968
XGBoost	0.9761	0.98	0.98	0.98	0.9982
MLP	0.9826	0.98	0.98	0.98	0.9987

5.3 Model Comparison and Analysis

Based on evaluation metrics, the models show varying strengths for CKD detection:

- **SVM (RBF)** achieved high performance with Accuracy 97.61%, macro Recall 0.98, and AUC-ROC 0.9981, demonstrating robustness in detecting CKD cases. Its high Recall is especially important in minimizing false negatives.
- **MLP** performed slightly better in Accuracy (98.26%) and AUC-ROC (0.9987), capturing complex non-linear relationships. However, it requires careful tuning and deeper architectures, which may impact interpretability.
- **XGBoost and Random Forest** performed similarly, with accuracies around 97–98% and high AUCs (0.9982 and 0.9968, respectively), providing reliable predictions and interpretability via feature importance.
- **Decision Tree** showed solid performance (Accuracy 96.31%, AUC 0.9809), but is less accurate than ensemble models.
- **Logistic Regression** offered good interpretability with moderate performance (Accuracy 83.30%), but lower Recall makes it less ideal for early CKD detection.

Trade-offs: SVM (RBF) and MLP offer the best detection capability but require more computational resources. Tree-based models provide interpretability and robustness. Logistic Regression remains useful for insights but may miss critical cases.

5.4 Conclusion and Model Selection

Considering the priority on Recall to minimize false negatives, **SVM (RBF)** is selected as the best model for CKD detection. Its combination of high Recall (0.98), Accuracy (97.61%), and AUC-ROC (0.9981) ensures that most CKD cases are correctly identified, supporting timely intervention by healthcare professionals. While MLP shows slightly higher Accuracy and AUC, SVM's balance of performance and interpretability makes it most suitable for deployment in clinical settings.

6. Ethical Considerations and Bias Mitigation

6.1 Potential Sources of Bias

The dataset, although synthetic, may introduce several forms of bias. **Data bias** may arise if the synthetic generation process underrepresents certain demographic groups such as elderly or minority populations. **Sampling bias** could occur if the dataset skews toward specific age ranges or health conditions, reducing generalizability. **Labeling bias** may also be present, as Chronic Kidney Disease (CKD) classification relies on predefined clinical thresholds that might not fully capture individual variability.

6.2 Ethical Principles and Mitigation Strategies

To ensure responsible and equitable AI development, we considered key ethical principles throughout the project:

- **Fairness:** We evaluated model performance across demographic subgroups (age and diabetes status) using demographic parity. The observed disparity in predictive accuracy was below 5%, indicating fair and consistent outcomes across groups. This reduces the risk of systematic disadvantage to any subgroup.
- **Transparency and Explainability:** We applied **SHAP (SHapley Additive exPlanations)** to interpret feature importance, allowing users and clinicians to understand which attributes (e.g., serum creatinine, blood pressure) most strongly influenced CKD predictions. This enhances trust and accountability in clinical decision support systems.
- **Accountability:** A disclaimer was included stating: *“This AI model is an assistive tool only—final medical decisions should be made by qualified healthcare professionals.”* This clarifies the scope of responsibility and prevents overreliance on algorithmic outputs.
- **Beneficence and Non-Maleficence:** Our model design prioritized **recall (sensitivity)** to minimize false negatives—ensuring that potential CKD patients are not overlooked. This ethical trade-off favors patient safety, aligning with medical best practices to avoid harm.
- **Privacy:** The dataset used was entirely anonymized and synthetic, containing no personally identifiable information. This safeguards patient confidentiality and complies with data protection principles such as GDPR.

Overall, these measures collectively promote fairness, transparency, accountability, and safety in applying AI for healthcare prediction tasks.

7. Reflections and Lessons Learned

7.1 Technical Reflections

The project highlighted the critical role of data quality in achieving reliable predictions. Through preprocessing steps such as normalization, handling missing values, and dimensionality reduction via PCA, model performance improved by approximately 10–15%. Evaluation results reinforced the importance of emphasizing **recall** over raw accuracy in healthcare applications, as detecting all possible CKD cases is ethically and clinically more valuable than minimizing false positives.

7.2 Teamwork and Project Management Reflections

Our team collaborated effectively using shared notebooks and regular virtual meetings. Clear task allocation enhanced productivity, while Git-based version control helped manage code conflicts and streamline integration. One key challenge was coordinating multiple updates to the same model files, which we resolved by establishing a consistent workflow and merge review process. This experience strengthened our communication, problem-solving, and project management skills.

7.3 Future Improvements

1. Source and validate the model on a real-world, diverse CKD dataset to assess generalizability.
2. Implement **SMOTE (Synthetic Minority Oversampling Technique)** to further address class imbalance.
3. Integrate additional explainability tools such as **LIME** to complement SHAP and enhance interpretability.
4. Develop a **web-based demonstration interface** for real-time CKD risk prediction and visualization.

Finally, this project emphasized that building AI for healthcare is not merely a technical challenge but also an ethical responsibility. We learned that prioritizing fairness, transparency, and patient safety is essential to ensure AI systems truly benefit society.

8. References

- [1] Kaggle, “Kidney Disease Risk Dataset,” *Kaggle*, 2025. [Online]. Available: <https://www.kaggle.com/datasets/miadul/kidney-disease-risk-dataset>
- [2] World Health Organization, “Chronic Kidney Disease Facts,” *World Health Organization*, 2023. [Online]. Available: <https://www.who.int/news-room/fact-sheets/detail/chronic-kidney-disease>
- [3] Scikit-learn Developers, “Scikit-learn: Machine Learning in Python,” *Scikit-learn*, 2025. [Online]. Available: <https://scikit-learn.org>
- [4] T. Chen and C. Guestrin, “XGBoost: A scalable tree boosting system,” in *Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining (KDD)*, 2016, pp. 785–794.
- [5] S. M. Lundberg and S. I. Lee, “A unified approach to interpreting model predictions,” in *Advances in Neural Information Processing Systems (NIPS)*, 2017, pp. 4765–4774.